

		II		
		c	s	
I	C	2 1	0	
	S	0	1 2	
	B	4 -1	1 3	

Figure 3.5 Battle of sexes game (left) and a 3×2 game (right) which is the same game with an extra strategy B for player I.

A mixed strategy equilibrium can exist also in 2×2 games that have pure strategy equilibria. As an example, consider the battle of sexes game, shown on the left in figure 3.5, which has the pure strategy equilibria (C, c) and (S, s) . As indicated by the boxes, the best response of the other player depends on the player's own strategy: If player I plays C , then player II's best response is c , and the best response to S is s . Suppose now that player I plays a mixed strategy, playing C with probability $1 - p$ and S with probability p . Clearly, when p is close to zero so that player I almost certainly chooses C , the best response will still be c , whereas if p is close to one the best response will be s . Consequently, there should be some probability so that player II will be indifferent. This probability p is found as follows. Given p , which defines the mixed strategy of player I, the expected payoff to player II when she plays c is $2(1 - p)$, and when she plays s that expected payoff is simply p . She is indifferent between c and s if and only if these expected payoffs are equal, that is, $2(1 - p) = p$ or $2 = 3p$, that is, $p = 2/3$. This mixed strategy of player I, where he plays C and S with probabilities $1/3$ and $2/3$, respectively, gives player II for both her strategies, the two columns, the same expected payoff $2/3$. Then player II can mix between c and s , and a similar calculation shows that player I is indifferent between his two strategies if player II uses the mixed strategy $(2/3, 1/3)$, which is the vector of probabilities for her two strategies c, s . Then and only then player I gets the same expected payoff for both rows, which is $2/3$. This describes the mixed equilibrium of the game, which we write as $((1/3, 2/3), (2/3, 1/3))$, as a pair (x, y) of mixed strategies x and y , which are probability vectors.

In short, the rule to find a mixed strategy equilibrium in a 2×2 game is to make the other player indifferent, because only in that case the other player can mix. This indifference means the expected payoffs for the two opponent strategies have to be equal, and the resulting equation determines the player's own probability.

The mixed strategy probabilities in the battle of sexes game can be seen relatively quickly by looking at the payoff matrices: For example, player II must give twice as much weight to strategy c compared to s , because player I's strategy C only gets one payoff unit from c whereas his strategy S gets two payoff units when player II chooses s . Because the strategy

pairs (C, s) and (S, c) give payoff zero, the two rows C and S give the same expected payoff only when c has probability $2/3$ and s has probability $1/3$.

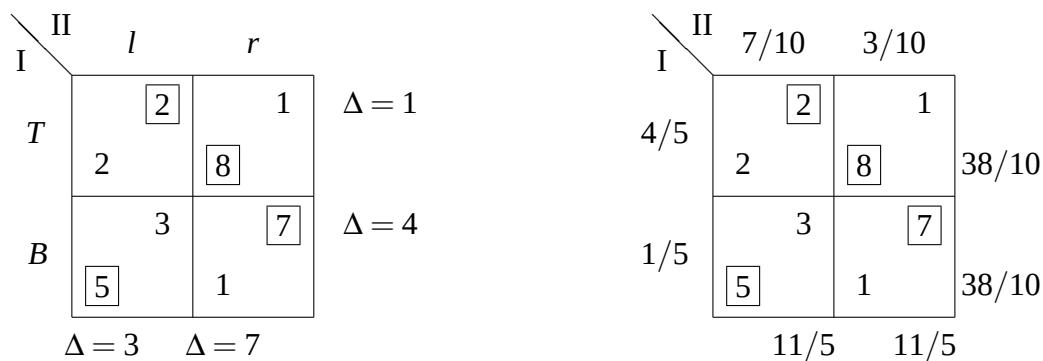


Figure 3.6 The “difference trick” to find equilibrium mixed strategy probabilities in a 2×2 game. The left figure shows the game and the difference in payoffs to the other player for each strategy. As shown on the right, these differences are assigned to the respective other own strategy and are re-normalised to become probabilities. The fractions $11/5$ and $38/10$ are the resulting equal expected payoffs to player II and player I, respectively.

We now explain a quick method, the “difference trick”, to find the probabilities in a mixed strategy equilibrium of a general 2×2 game. Consider the game on the left in figure 3.6. When player II plays T , then l is a best response, and when player II plays B , then r is a best response. Consequently, there must be a way to mix T and B so that player II is indifferent between l and r . We consider now the difference Δ in payoffs to the other player for both rows: When player I plays T , then the difference between the two payoffs to player II is $\Delta = 2 - 1 = 1$, and when player I plays B , then that difference, in absolute value, is $\Delta = 7 - 3 = 4$, as shown on the side of the game. (Note that these differences, when considered as differences between the payoffs for l versus r , have opposite sign because l is preferred to r against T , and the other way around against B ; otherwise, player II would always prefer the same strategy and could not be made indifferent.) Now the payoff difference Δ is assigned as a probability weight to the respective *other* strategy of the row player, meaning T is given weight 4 (which is the Δ computed for B), and B is given weight 1 (which is the Δ for T). The probabilities for T and B are then chosen proportional to these weights, so one has to divide each weight by 5 (which is the sum of the weights) in order to obtain probabilities. This is shown on the right in figure 3.6. The expected payoffs to player II are also shown there, at the bottom, and are for l given by $(4 \cdot 2 + 1 \cdot 3)/5 = 11/5$, and for r by $(4 \cdot 1 + 1 \cdot 7)/5 = 11/5$, so they are indeed equal as claimed. Similarly, the two payoff differences for player I in columns l and r are 3 and 7, respectively, so l and r should be played with probabilities that are proportional to 7 and 3, respectively. With the resulting probabilities $7/10$ and $3/10$, the two rows T and B get the same expected payoff $38/10$. A proof that this “difference trick” always works is the subject of exercise 3.2.

Next, we consider games where one player has two and the other more than two strategies. The 3×2 game on the right in figure 3.5 is like the battle of sexes game, except that player I has an additional strategy B . Essentially, such a game can be analysed by considering the 2×2 games obtained by restricting both players to two strategies only, and checking if the resulting equilibria carry over to the whole game. First, the pure strategy equilibrium (C, c) of the original battle of sexes game is also an equilibrium of the larger game, but (S, s) is not because S is no longer a best response to s because B gives a larger payoff to player I. Second, consider the mixed strategy equilibrium of the smaller game where player I chooses C and S with probabilities $1/3$ and $2/3$ (and B with probability zero), so that player II is indifferent between c and s . Hence, the mixed strategy $(2/3, 1/3)$ of player II is still a best response, against which C and S both give the same expected payoff $2/3$. However, this is not enough to guarantee an equilibrium, because C and S have to be best responses, that is, their payoff must be at least as large as that for the additional strategy B . That payoff is $(-1) \cdot 2/3 + 3 \cdot 1/3 = 1/3$, so it is indeed not larger than the payoff for C and B . In other words, the mixed equilibrium of the smaller game is also a mixed equilibrium of the larger game, given by the pair of mixed strategies $((1/3, 2/3, 0), (2/3, 1/3))$.

Other “restricted” 2×2 games are obtained by letting player I play only C and B , or only S and B . In the first case, the difference trick gives the mixed strategy $(3/5, 2/5)$ of player II (for playing c, s) so that player I is indifferent between C and B , where both strategies receive expected payoff $3/5$. However, the expected payoff to the third strategy S is then $4/5$, which is higher, so that C and B are not best responses, which means we cannot have an equilibrium where only C and B are played with positive probability by player I. Another reason why no equilibrium strategy of player II mixes C and B is that against both C and B , player II’s best response is always l , so that player I could not make player II indifferent by playing in that way.

Finally, consider S and B as pure strategies that player I mixes in a possible equilibrium. This requires, via the difference trick, the mixed strategy $(0, 3/4, 1/4)$ of player I (as probability vector for playing his three pure strategies C, S, B) so that player II is indifferent between c and s , both strategies receiving payoff 1 in that case. Then if player II uses the mixed strategy $(1/2, 1/2)$ (in order to make player I indifferent between S and B), the expected payoffs for the three rows C, S, B are $1/2$, 1, and 1, respectively, so that indeed player I only uses best responses with positive probability, which gives a third Nash equilibrium of the game.

Why is there no mixed equilibrium of the 3×2 game in figure 3.5 where player I mixes between his three pure strategies? The reason is that player II has only a single probability at her disposal, which does not give enough freedom to satisfy two equations, namely indifference between C and S as well as between S and B (the third indifference between C and B would then hold automatically). Namely, we have already computed the probabilities $(2/3, 1/3)$ for c, s that are needed to give the same expected payoff for C and B , against which the payoff for row B is different, namely $1/3$. This alone suffices to show that it is not possible to make player I indifferent between all three pure strategies, so there is no equilibrium where player I mixes between all of them. It certain games, which are called “degenerate” and which are treated in section 3.12, it is indeed the case that player II, say, mixes only be-

tween two pure strategies but where “by accident” three strategies of player I have the same optimal payoff. This is normally not the case, and leads to complications when determining equilibria, which will be discussed in the later section on degenerate games.

3.11 The upper envelope method

The topic of this section is the “upper envelope” diagram that simplifies mixed finding equilibria, in particular of games where one player has only two strategies.

The left graph in figure 3.7 shows the upper envelope diagram for the left game in figure 3.5. This diagram is a plot of the expected payoff of one player against the mixed strategy of the other player. We use this typically when that mixed strategy is determined by a single probability, that is, we consider a player who has only two pure strategies, in this example player II. The horizontal axis represents the probability for playing the second pure strategy s of player II, so this is $q = \text{prob}(s)$. In the vertical direction, we plot the resulting expected payoffs to the other player, here player I, for his pure strategies. In this game, this is the expected payoff $1 - q$ when he plays row C , and the expected payoff $2q$ when he plays row S . The plots are lines because expected payoffs are linear functions of the mixed strategy probabilities.

These lines are particularly easy to obtain graphically. To see this, consider row C of player I. When player II plays her left strategy c (where $q = 0$), row C gives payoff 1 (according to the table of payoffs to player I). When player II plays her right strategy s (where $q = 1$), row C gives payoff 0. The probabilities $q = 0$ or $q = 1$ and corresponding payoffs define the two endpoints with coordinates $(0, 1)$ and $(1, 0)$ of a line that describes the expected payoff $1 - q$ as a function of q , which consists of the points $(q, 1 - q)$ for $q \in [0, 1]$. As explained in section 3.7 above, this line is just the set of convex combinations $(1 - q) \cdot (0, 1) + q \cdot (1, 0)$ of the two endpoints. The first component of such a convex combination is always q and second component $(1 - q) \cdot 1 + q \cdot 0$ is the expected payoff. For row S , this expected payoff is $(1 - q) \cdot 0 + q \cdot 2$, so the line of expected payoffs for S connects the left endpoint $(0, 0)$ with the right endpoint $(1, 2)$.

The upper envelope diagram can also be described as the “goal-post” method. First, identify a player who has only two pure strategies (here player II). Second, plot the probability (here q) for the second of these pure strategies horizontally, along the interval $[0, 1]$. Third, erect a vertical line (a “goal-post”) at the left endpoint 0 and at the right endpoint 1 of that interval. Forth, do the following for each pure strategy of the other player (which are here rows C and S for player I): Mark the payoff (as given in the game description) against the left strategy (of player II, that is, against $q = 0$) as a “height” on the left goal-post, and against the right strategy ($q = 1$) as a height on the right goal-post, and connect the two heights by a line, which gives the expected payoff as a function of q for $0 \leq q \leq 1$.

Note that for these goal-posts it is very useful to consider player II’s mixed strategy as the vector of probabilities $(1 - q, q)$ (according to the second step above), because then the left goal-post corresponds to the left strategy of player II, and the right goal-post to the right

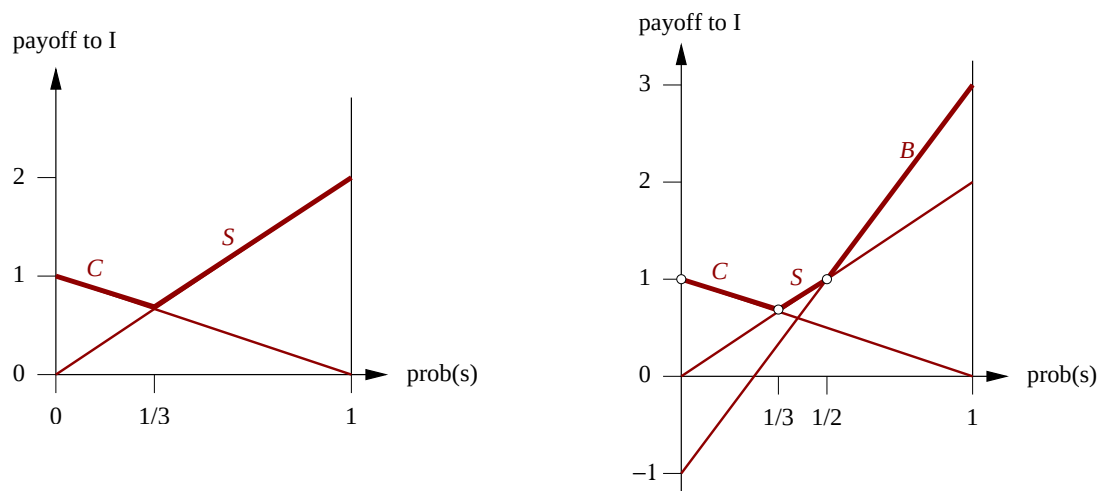


Figure 3.7 Upper envelope of expected payoffs to player I for the two games in figure 3.5.

strategy of player II. (This would not be the case if q was taken as the probability for the left strategy, writing player II's mixed strategy as $(q, 1 - q)$, because then $q = 0$ would be the left goal-post but represent the right pure strategy of player II, creating an unnecessary source of confusion.)

In the left diagram in figure 3.7, the two lines of expected payoffs for rows C and S obviously intersect exactly where player I is indifferent between these two rows, which happens when $q = 1/3$. However, the diagram not only gives information about this indifference, but about player I's preference in general: Any pure strategy that is a *best response* of player I is obviously the *topmost* line segment of all the lines describing the expected payoffs. Here, row C is a best response whenever $0 \leq q \leq 1/3$, and row S is a best response whenever $1/3 \leq q \leq 1$, with indifference between C and S exactly when $q = 1/3$. The *upper envelope* of expected payoffs is the *maximum* of the lines that describe the expected payoffs for the different pure strategies, and it is indicated by the bold line segments in the diagram. Marking this upper envelope in bold is therefore the fifth step of the “goal-post” method.

The upper envelope is of particular use when a player has more than two pure strategies, because mixed equilibria of a 2×2 are anyhow quickly determined with the “difference trick”. Consider the upper envelope diagram on the right of figure 3.7, for the 3×2 game in figure 3.5. The extra strategy B of player I gives the line connecting height -1 on the left goal-post to height 3 on the right goal-post, which shows that B is a best response for all q so that $1/2 \leq q \leq 1$, with indifference between S and B when $q = 1/2$. The upper envelope consists of three line segments for the three pure strategies C, S, and B of player I. Moreover, there are only two points where player I is indifferent between any two such strategies, namely between C and S when $q = 1/3$, and between S and B when $q = 1/2$. These two points, indicated by small circles on the upper envelope, give the only two mixed strategy probabilities of player II where player I can mix between two pure strategies (the small circle for $q = 0$ corresponds to the pure equilibrium strategy c). There is a third indifference between C and B, for $q = 2/5$, where the two lines for C and B intersect, but this intersection

point is below the line for S and hence not on the upper envelope. Consequently, it does not have to be investigated as a possible mixed strategy equilibrium strategy where player I mixes between the two rows C and B because they are not best responses.

Given these candidates of mixtures for player I, we can quickly find if and how the two respective pure strategies can be mixed in order to make player II indifferent using the difference trick, as described in the previous section.

So the upper envelope restricts the pairs of strategies that need to be tested as possible strategies that are mixed in equilibrium. Exercise 3.3, for example, gives a 2×5 game, where the upper envelope may consist of up to five line segments (there may be fewer line segments if some pure strategy of player II is never a best response). Consequently, there are up to four points where these line segments join up, so that player II is indifferent between two strategies and can mix. Without the upper envelope, there would not be four but ten (two out of five) pure strategy pairs to be checked as possible pure strategies that player II mixes in equilibrium, which would be much more laborious to investigate.

In principle, the upper envelope diagram is also defined for the expected payoffs against a mixed strategy that mixes between more than two pure strategies. This diagram is much harder to draw and visualise in this case because it requires higher dimensions. We demonstrate this for the game on the right in figure 3.5, although we do not suggest to use this as a method to find all equilibria of such a game.

Consider the set of mixed strategies of player I for the 3×2 game of figure 3.5, which is a triangle. Then the payoff to the other player II needs an extra “vertical” dimension. A perspective drawing of the resulting three-dimensional picture is shown in figure 3.8, in several stages. First (top left), the mixed strategy triangle is shown with goal-posts erected on its corners $(1,0,0)$, $(0,1,0)$, and $(0,0,1)$. Next (top right), we mark the heights 2, 0, and 4 on these posts which are the payoffs to player II for her first strategy c . The resulting expected payoffs define a plane above the triangle through these height markings. The third picture (bottom left) does the same for the second strategy s of player II, which has payoffs 0, 1, and 1 as heights in the three corners of the triangle. The final picture (bottom right) gives the upper envelope as the maximum of the two expected-payoff planes (which in the lower dimension used to be drawn in bold). The small circles indicate the three mixed strategies of player I which are his strategies, respectively, in the three Nash equilibria $((1,0,0), (1,0))$, $((1/3, 2/3, 0), (2/3, 1/3))$, and $((0, 3/4, 1/4), (1/2, 1/2))$. The first of these three equilibria is the pure strategy equilibrium (C, c) , and the last two involve mixing both columns c and s of player II. Hence, player II must be indifferent between both c and s , so player I must choose a mixed strategy where the planes of expected payoffs for these two strategies c and s of player II meet. (These two planes meet in a line, which is given by any convex combination of the two mixed strategies $(1/3, 2/3, 0)$ and $(0, 3/4, 1/4)$ of player I; however, player I must choose one of these two endpoints because he can only mix either between C and S or between S and B as described earlier, depending on the mixed strategy of player II.)